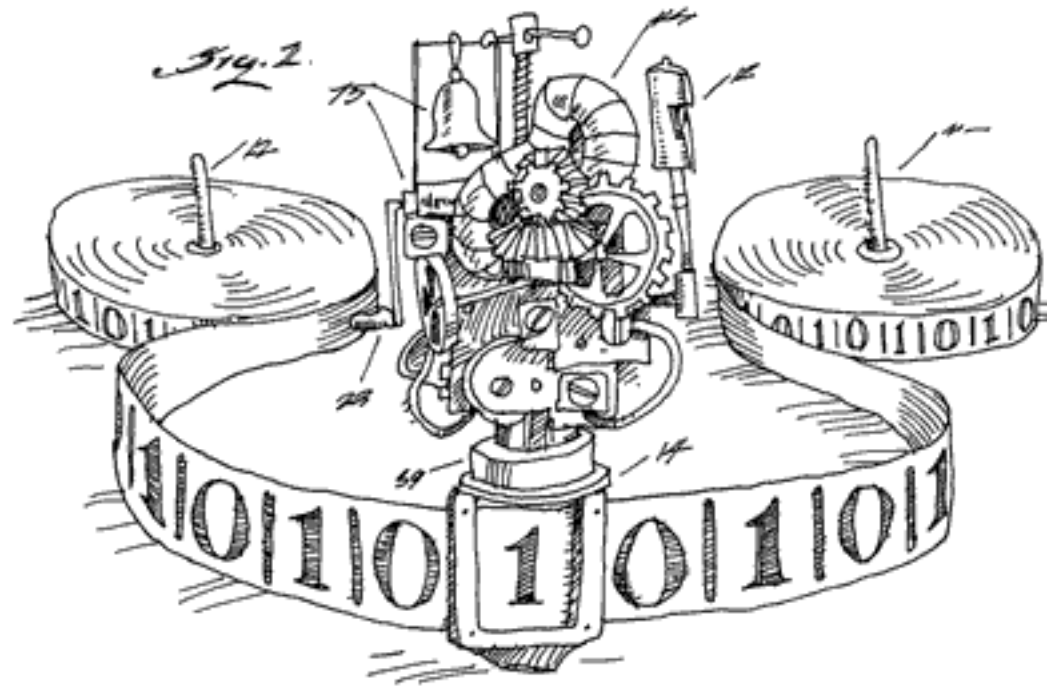


CSC4010: Introduction to Theoretical Computer Science



Lecture 8

Adapted from Sipser's MIT OCW Course on TOC

Last Time

- Is there an explicit computational problem which cannot be solved in any computational problem?
- We showed that L_{self} , and A_{TM} are undecidable via diagonalization.
- A language L is decidable if and only if L and $\bar{L}$ are both recognizable. We used this to conclude that $\bar{A}_{TM}$ is undecidable.
- We showed HALT_{TM} is undecidable by reducing A_{TM} to HALT_{TM} .
- **Proof Strategy:** If there is a decider H for HALT_{TM} , then we can construct a decider M' for A_{TM} . But A_{TM} is undecidable.

Reducibility Method: Undecidability of HALT_{TM}

- Let $\text{HALT}_{TM} = \{ \langle M, w \rangle \mid M \text{ halts on } w \}$.

Theorem: HALT_{TM} is undecidable.

- Proof:** Proof by contradiction. Assume there exists H deciding HALT_{TM} .

Idea: Use H to construct a Turing Machine M' deciding A_{TM} .

$M' =$ “On input $\langle M, w \rangle$

1. Simulate H on input $\langle M, w \rangle$.
2. Reject if H rejects.
3. If H accepts, run M on w until it halts.
 - (a) If M accepts, then accept.
 - (b) If M rejects, then reject.”

Claim: If H exists, then M' decides A_{TM} .

Reducibility - Concept

- If we have two languages (or problems) A and B , then A being reducible to B implies we can use B to solve A .
- **Example:** Measuring the area of a rectangle is reducible to measuring the lengths of its sides.
- If A is reducible to B , then solving B gives a solution to A .
 - Then, B is easy implies A is easy.
 - Then, A is hard implies B is hard.

This is the form we will use.

Another Example: EMPTY_{TM}

- Let $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$.

Theorem: EMPTY_{TM} is undecidable.

- Proof:** Proof by contradiction. Assume there exists R deciding EMPTY_{TM} .

Idea: Use R to construct a Turing Machine M' deciding A_{TM} .

Attempt 1: Given input $\langle M, w \rangle$ for M' , we pass $\langle M \rangle$ through R .

Another Example: EMPTY_{TM}

- Let $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$.

Theorem: EMPTY_{TM} is undecidable.

- Proof:** Proof by contradiction. Assume there exists R deciding EMPTY_{TM} .

Idea: Use R to construct a Turing Machine M' deciding A_{TM} .

Attempt 2: Given input $\langle M, w \rangle$ for M' , transform M to M_w defined as follows:

$M_w =$ "On input $x \in \{0,1\}^*$

1. If $x \neq w$, then reject.
2. Simulate M on w .

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

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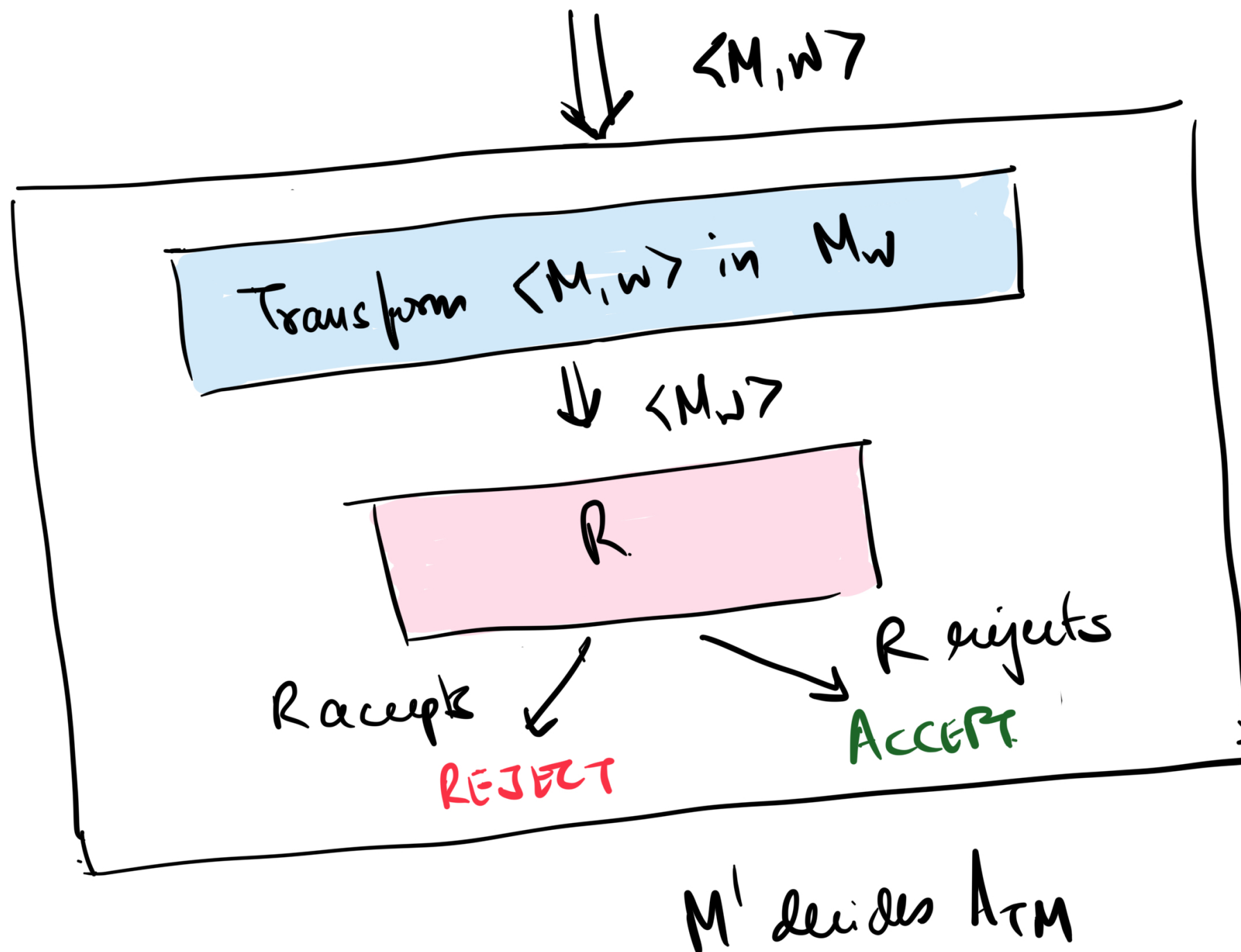
1. Transform $\langle M, w \rangle$ into M_w .
2. Run R on $\langle M_w \rangle$.
3. **Accept** if R **rejects**.
4. **Reject** if R **accepts**.

$M_w =$ "On input $x \in \{0,1\}^*$

1. If $x \neq w$, then reject.
2. Simulate M on w .

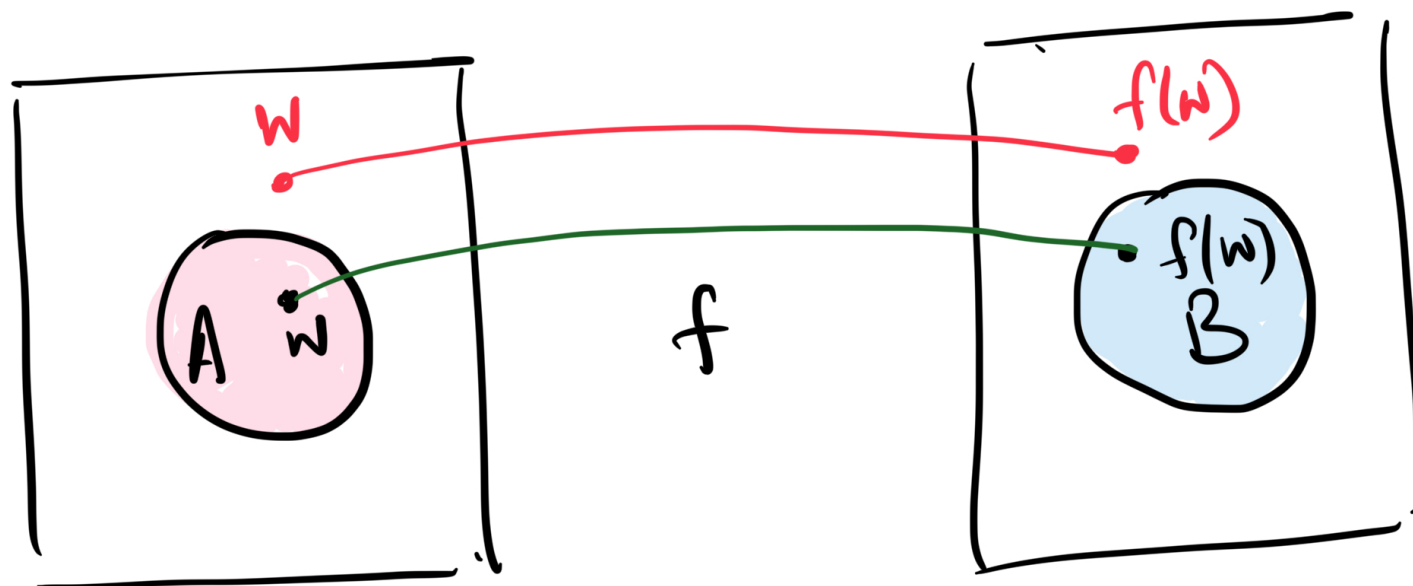
$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

Another Example: EMPTY_{TM}



Mapping Reducibility

- **Definition:** A function $f : \Sigma^* \rightarrow \Sigma^*$ is said to be **computable** if there exists a Turing Machine F , which on input $w \in \Sigma^*$ terminates with just $f(w)$ written on its tape, for any w .
- **Definition:** A language A is **mapping reducible** to a language B (denoted $A \leq_m B$) if and only if there exists a computable function $f : \Sigma^* \rightarrow \Sigma^*$ such that for every $w \in \Sigma^*$, $w \in A \iff f(w) \in B$.



f is called a mapping reduction from A to B .

$$A_{TM} \leq_m \overline{\text{EMPTY}}_{TM}$$

Solution: Given input $\langle M, w \rangle$ for A_{TM} , $f(\langle M, w \rangle) = \langle M_w \rangle$ defined as follows:

$M_w =$ "On input $x \in \{0,1\}^*$

1. If $x \neq w$, then reject.
2. Simulate M on w .

Claim: $f(\langle M, w \rangle) \in \overline{\text{EMPTY}}_{TM} \iff \langle M, w \rangle \in A_{TM}$.

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

Mapping Reducibility: Properties

Theorem: If $A \leq_m B$ and B is decidable, then A is decidable.

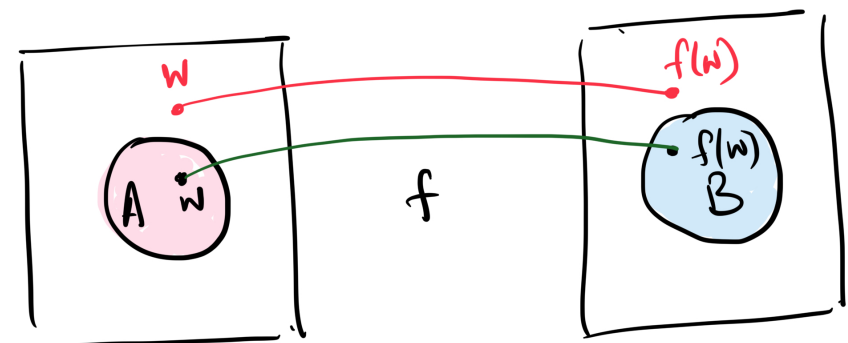
“If B is at least as hard as A , and B is decidable then so is A .”

- **Proof:** Let M_B be a decider for B . We build a decider for M_A for A as follows:

M_A = “On input $w \in \{0,1\}^*$

1. Compute $f(w)$.
2. Run M_B on $f(w)$ and output its answer.

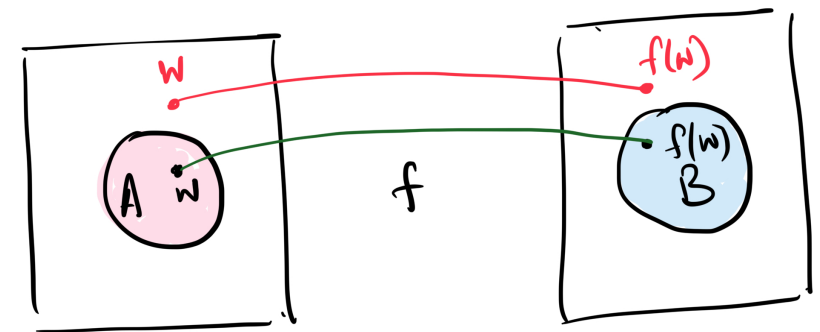
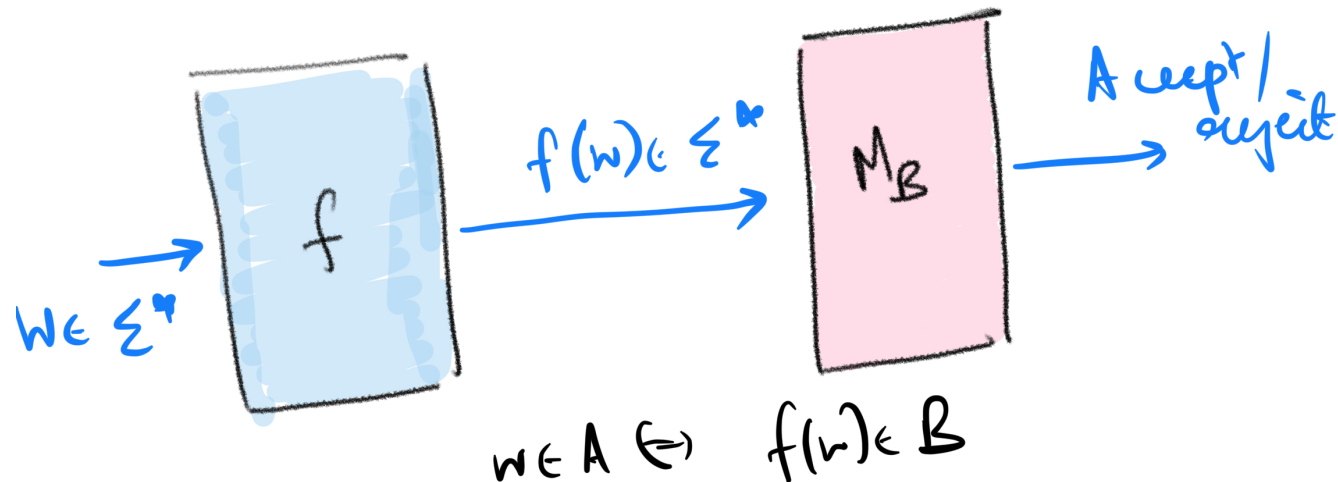
M_A accepts $w \iff M_B$ accepts $f(w)$
 $\iff f(w) \in B$
 $\iff w \in A$.



Mapping Reducibility: Properties

Theorem: If $A \leq_m B$ and B is decidable, then A is decidable.

A decider M_A for A :

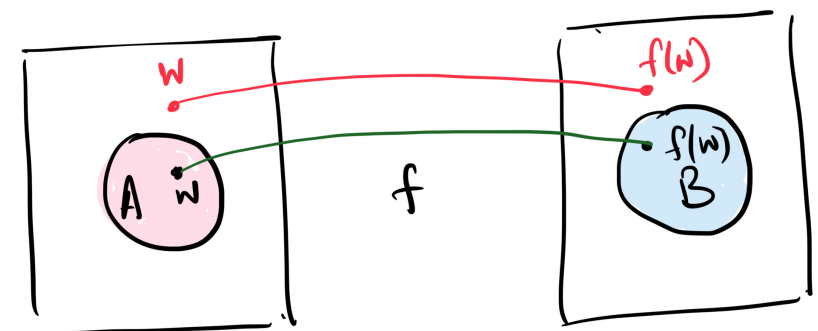
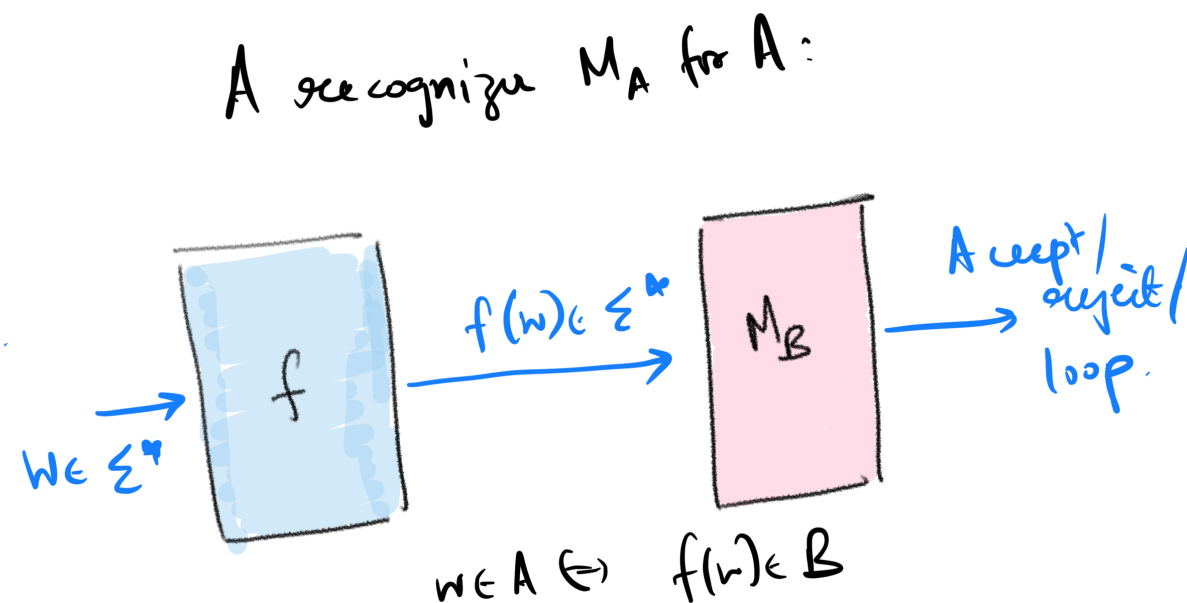


Corollary: If $A \leq_m B$ and A is undecidable, then B is undecidable.

A recipe for undecidability: To show B is undecidable, find undecidable A and a mapping reduction from A to B .

Mapping Reducibility: Properties

Theorem: If $A \leq_m B$ and B is recognizable, then A is recognizable.

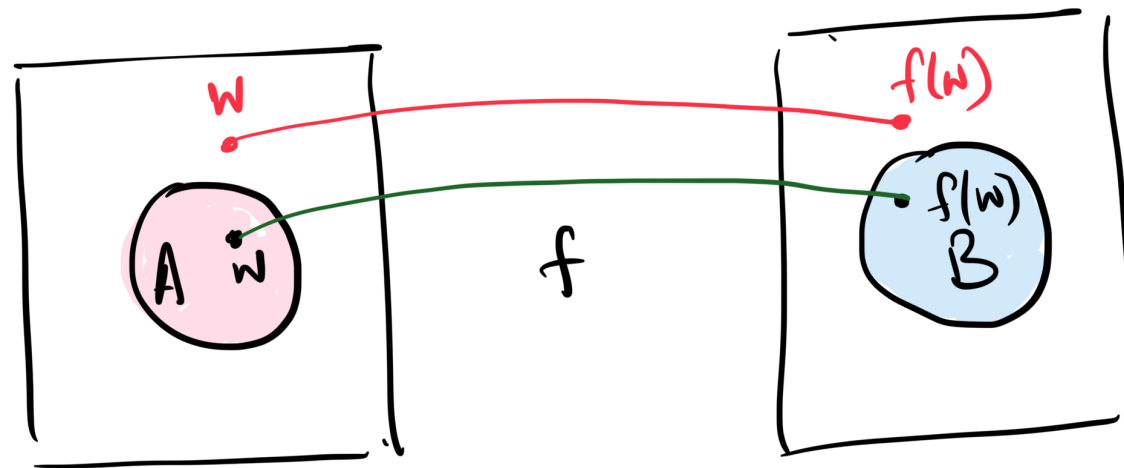


Corollary: If $A \leq_m B$ and A is unrecognizable, then B is unrecognizable.

A recipe for unrecognizability : To show B is unrecognizable, find unrecognizable A and a mapping reduction from A to B .

Mapping Reducibility: Properties

Theorem: If $A \leq_m B$ then $\bar{A} \leq_m \bar{B}$.



Using Mapping Reduction for Unrecognizability

EMPTY_{TM} is unrecognizable

- Let $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$.

Theorem: EMPTY_{TM} is undecidable.

- Proof:** Proof by contradiction. We will show $\bar{A}_{TM} \leq_m \text{EMPTY}_{TM}$.

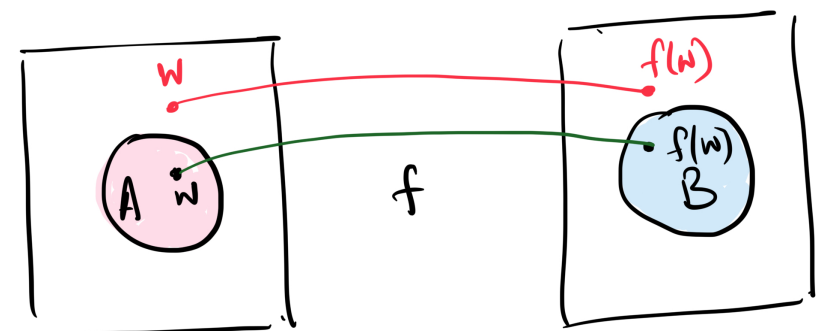
Reduction Function: $f(\langle M, w \rangle) = \langle M_w \rangle$

$M_w =$ "On input $x \in \{0,1\}^*$

1. If $x \neq w$, then reject.
2. Simulate M on w .

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

$$\begin{aligned} f(\langle M, w \rangle) \in \text{EMPTY}_{TM} &\iff \langle M_w \rangle \in \text{EMPTY}_{TM} \\ &\iff L(M_w) = \emptyset \\ &\iff M \text{ does not accept } w \\ &\iff \langle M, w \rangle \in \bar{A}_{TM} \end{aligned}$$



Reducibility Terminology

Why do we use the term “reduce”?

- When we reduce A to B , we show how to solve A by using B . We conclude that A is no harder than B . (This suggests the $\leq_m$ notation).
- Possibility 1: We bring A 's difficulty down to B 's difficulty.
- Possibility 2: We bring B 's difficulty up to A 's difficulty.

Summary

- Introduced the mapping reducibility to show undecidability/unrecognizability.
- Showed EMPTY_{TM} is undecidable and unrecognizable.
- Next time: more examples.